MILITARY SPECIFICATION

IGNITER, ROCKET MOTOR, MARK 297 MOD 1

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the MARK 297 MOD 1 rocket motor igniter referred to herein as the igniter.

1.2 Classification. The igniters shall be of the following types:

Type I	Delivery units
Type II	Static test units for lot acceptance

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-763	Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resistant.

MILITARY

MIL-A-47284	Adhesive, Epoxy Resin Base.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging and Packing Levels.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-810	Environmental Test Methods.
MIL-STD-45662	Calibration Systems Requirements.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

639AS4612	Igniter, Rocket Motor.
1204AS204	Spring, Constant Force.
1204AS205	Shaft.
1204AS215	Seal.
1204AS211	Rupture Disk Assembly.
639AS4601	Plug, Igniter Tube.
MS-33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal.
MS-51957	Screw, Machine-Pan Head, Cross-Recessed, Corrosion-Resisting Steel.
AMS-5653	Bars, Forgings, Tubing and Rings.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178 Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS.

3.1 Item description. The igniter consists of a safe-arm assembly, closure, and a tube containing the reduced smoke igniter propellant charge. Other components are as identified on Drawing 639AS4612.

3.2 Characteristics.

3.2.1 Arm function performance. When functioned at 25 ±3 degrees Celsius (°C) in the armed position, the Type II igniter assembly shall meet the requirements of Figure 1. When visually inspected after firing, the general appearance of the igniter shall indicate no abnormal conditions and shall show no indication of gas flow other than through the twelve exhaust ports.

3.2.2 Safe function. When the Type I igniter is fired in the safe position, the seal (Drawing 1204AS215), boron-potassium-nitrate pellets, and rupture disk (Drawing 1204AS211) shall show no visual evidence of hot gas leakage. After functioning in the safe position, the arming rod shall withstand an arming torque of 100 inch-pounds minimum in the counter-clockwise direction without rotation.

3.2.3 Ignition circuit resistance. The igniter squib bridge resistance shall be 0.5 ±0.1 ohm at 21 ±6°C.

3.3 Design and construction.

3.3.1 Type I. The rocket motor igniter shall be fabricated and assembled in accordance with Drawing 639AS4612 and associated documents.

3.3.2 Type II. The Type II igniter shall be fabricated and assembled the same as the Type I igniter except that the constant force spring (Drawing 1204AS204), shaft (Drawing 1204AS205), and screw (Drawing MS-51957) are not required, and the igniter tube plug (Drawing 639AS4601) shall be replaced with a special test plug as shown on Figure 2.

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3.3.3 Facilities. The contractor shall provide facilities and working environments which will assure compliance with the design, construction, and performance requirements herein.

3.4 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2), a first article sample consisting of 22 igniters shall be delivered to the testing activity designated in the contract or purchase order (see 6.2). The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample is at the risk of the contractor.

3.5 Workmanship. The igniter and all components and subassemblies shall be of uniform, high quality workmanship. The parts and assemblies shall be clean and free from foreign materials, metal particles, hydrocarbons, explosive material (on external surfaces), or other contaminates.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspections (see 4.4)

b. Quality conformance inspections (see 4.5)

4.3 Inspection conditions. Unless otherwise specified herein, the igniter shall be tested at the atmospheric pressure conditions of the test facility.

4.3.1 Equipment and facilities. Test equipment and inspection facilities shall be provided by the testing activity, and shall be of sufficient quality and quantity to permit the performance of the required tests and measurements under the conditions, and to the accuracies specified herein.

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4.3.2 Maintenance and calibration. All test equipment shall be calibrated and maintained in accordance with the provisions of MIL-STD-45662.

4.3.3 Test equipment. The test equipment shall provide a recording system having capabilities of measuring:

a. Pressure of 1,800 pounds per square inch gage (psig) with an accuracy of ±1.5 percent.

b. Response of 1,000 pounds per square inch per millisecond, minimum.

c. Time graduations of 10 ±2 milliseconds per inch.

4.4 First article inspection. First article samples shall be subjected to all of the inspections specified in Table I.

4.4.1 First article sample acceptance criteria. Failure of any igniter to pass any of the tests specifed herein shall cause rejection of the first article sample. Corrective action acceptable to the Government shall be required for all failures.

4.4.1.1 Retest. At the discretion of the procuring activity, preproduction tests, or any portion thereof, shall be repeated under any of the following conditions:

a. The manufacturer has modified his product through a change of lot of raw material, production procedures, or methods. The contractor shall notify the procuring activity prior to the incorporation of any such changes, and shall provide quantitative evidence of the effect of such changes on the performance or characteristics of the product. The requirement for retesting will be based on evaluation by the procuring activity of the evidence supplied.

b. Where there is evidence that the quality of the product has not been maintained. This evidence may be in the form of accumulated failure reports of the products, of system failures attributable to the product, or failure of the product to pass any of the tests for quality verification that may be conducted by or for the procuring activity.

c. Any change in design or documentation by the procuring activity.

d. Any interruption of production by a qualified contractor.

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TABLE I. First article inspection program.

Test	Para.	Igniter number																					
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Igniter type	3.3	II	II	II	II	II	I	I	I	I	I	I	II	II	II	II	II	I	I	I	I	I	I
Visual examination	4.6.6	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Circuit resistance	4.6.3	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B
Thermal cycle	4.6.5	C	C	C	C	C	C	C	C										C	C	C	C	
Vibration	4.6.4						D	D	D	C	C	C							D	D	D	D	
Circuit resistance	4.6.3	D	D	D	D	D	E	E	E	D	D	D	C										
Arm-function (25 ±3°C)	4.6.1	E	E	E	E	E			F	E	E	E	D	C	C	C	C						
Safe-function	4.6.2																						
Hot (63 ±3°C)							F											C	E	E			
Cold (-54 ±3°C)								F													E	E	C

NOTE 1. Letters indicate tests. These tests shall be performed in alphabetical sequence.

2. Igniters 8 through 11 will not have pressure-time performance measurements taken. Post-fire condition (visual) shall be comparable to remainder of first article igniters.

3. The pressure plug for Type II igniters shall be sealed during environmental testing.

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4.5 Quality conformance inspections. Each igniter in the lot shall be subjected to the tests of 4.6.3 and 4.6.6. Sample Group A (see 4.5.2) shall be subjected to the test of 4.6.1. Sample Group B (see 4.5.2) shall be subjected to the test of 4.6.2.

4.5.1 Lot. A production lot of igniters shall consist of igniters assembled in a continuous operation using the same equipment, processes, and procedures and presented for acceptance. An igniter lot shall include no more than one igniter grain lot.

4.5.2 Sampling. Arm and safe function test samples shall be randomly selected from the lot in accordance with Table II. The samples shall be divided into Groups A and B.

4.5.2.1 Acceptance criteria. Failure of any igniter to meet any of the requirements specified herein shall be cause for rejection of the lot represented.

TABLE II. Quality conformance sampling plan.

Lot size	Sample size	Sample distribution	
		Groups	
		A 1/	B 2/
Up to 800	30	24	6
801-3200	40	32	8

1/ Type II igniters.

2/ Type I igniters

4.6 Tests and examination.

4.6.1 Arm function performance. The pressure versus time function test shall be conducted utilizing the Type II igniter (see 3.3.2), which will interface with a pressure transducer as shown on Figure 2. The igniter shall be conditioned at 25 ±3°C for a minimum of 4 hours. Subsequent to conditioning, the igniter shall be connected to a pressure tranducer and a power supply and fired within 20 minutes of removal from conditioning. The power supply shall deliver 3.5 ±0.5 amps to the squib circuit consisting of the safe-arm unit housing (ground) and the spade terminal of the squib (input). The arming rod shall be rotated to the arm position and the unit functioned. To be acceptable, the igniter shall meet the requirements of 3.2.1.

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4.6.2 Safe function. Condition half of the sample units at 63 ±3°C and the other half of the sample units at -54 ±3°C for 4 hours minimum. The igniters shall then be fired in the safe position using a firing current of 3.5 ±0.5 amperes direct current within 20 minutes after removal from the conditioning chamber. After firing, a counterclockwise torque of 100 inch-pounds minimum shall be applied to the arming rod. The igniter shall then be disassembled for inspection of seal, boron-potassium-nitrate pellets, and rupture disk for evidence of hot gas leakage. To be acceptable, the igniter shall meet the requirements of 3.2.2.

4.6.3 Ignition circuit-resistance. The resistance of the igniter circuit shall be measured using a circuit tester that limits the current flow to 0.010 amperes maximum, and is accurate within 0.01 ohm. To be acceptable, the igniter shall meet the requirements of 3.2.3.

4.6.4 Vibration. The igniter (locked in the armed position) shall be subjected to random vibration, while conditioned at temperature extremes, in accordance with MIL-STD-810, Method 514 except as noted herein. The random vibration profile shall be a flat power spectral density from 10 hertz (Hz) to 2,000 Hz at a level of 0.02 gravity units squared per Hz. Vibration control filter cut-off characteristics shall roll-up to 10 Hz and roll-off at 2,000 Hz, equal to or greater than the rate of +18 decibel (dB) per octave and -18 dB per octave, respectively. The igniter shall utilize service attachment when assembled to the vibration fixture duplicating the rocket motor and warhead interfaces. The igniter, while assembled to the vibration fixture, shall be conditioned for a minimum of 4 hours prior to vibration at each of the three following temperatures: -54 ±3°C, 63 ±3°C, and 25 ±3°C. The igniter, while exposed to each of the above temperatures shall be subjected to random vibrations for a period of no less than 1 hour per axis for a total of 9 hours vibration. After completion of the vibration testing the igniter shall meet the requirements of 3.2 and 3.5.

4.6.5 Thermal cycle. The igniter specified herein shall be subjected to five full temperature cycles with the safe-arm assembly in the safe position. The temperature limits shall be -54 ±3°C and 63 ±3°C. A full cycle shall consist of conditioning the unit for 4 hours +30 minutes, -O minutes at one temperature limit, transferring the unit within 10 minutes to the other temperature limit, conditioning the unit for 4 hours +30 minutes, -O minutes and return to the original temperature limit. One-half of the sample shall begin cycling at -54°C and one-half at 63°C. After testing, each unit shall meet the requirements of 3.2 and 3.5.

4.6.6 Visual examination. The igniter shall be visually inspected for conformance to the workmanship, finish, cleanliness, presence of components, visual arm indication, and marking requirements.

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5. PACKAGING

5.1 Packaging and packing. Packaging and packing levels shall be in accordance with FED-STD-102.

5.1.1 Level C. The igniter shall be packaged and packed in accordance with normal commercial practice in such a manner as to prevent damage during shipping and storage. Packing shall conform to CFR 171-178.

5.2 Marking. The packaging and packing shall be marked in accordance with MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. The MARK 297 MOD 1 igniter is intended for use in rocket motors.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article sample. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article sample should consist of 22 igniters as specified in 3.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article. First article approvel is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

c. Responsibility for inspection, if other than as specified in 4.1.

6.3 Definitions.

6.3.1 Zero time (T_0). T_0 is defined as the time that the current is applied to the initiator.

6.3.2 Delay time (T_1). The time required from zero time (T_0) until pressure rise is indicated.

6.4 Safety precautions. The loading, assembly, and handling of the explosives, subassemblies , and finished items covered by this specification involve hazardous operations, and therefore require explosives

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safety precautions. Standard safety precautions for explosive-loaded items are contained in DOD 4145.26M, “DOD Contractor’s Safety Manual for Ammunition, Explosives and Related Dangerous Material.”

Preparing activity:
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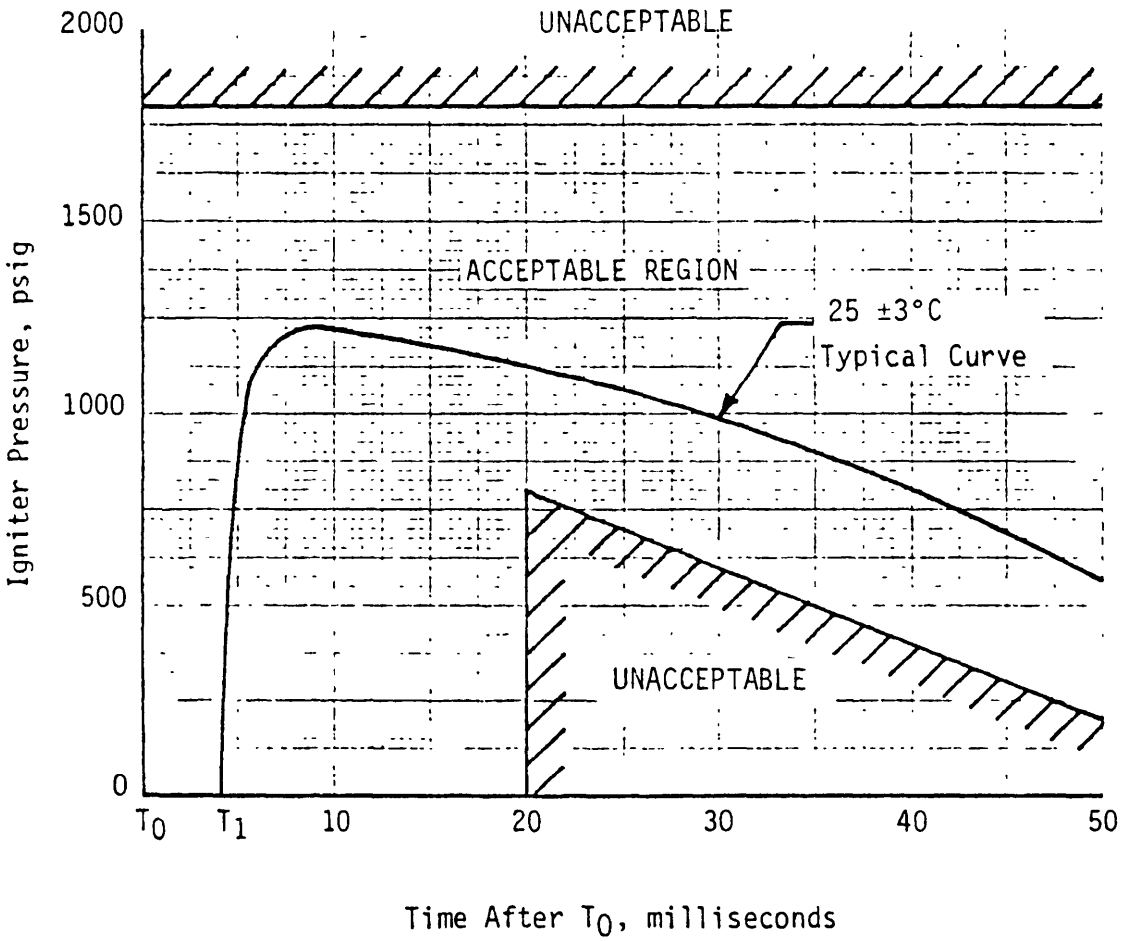
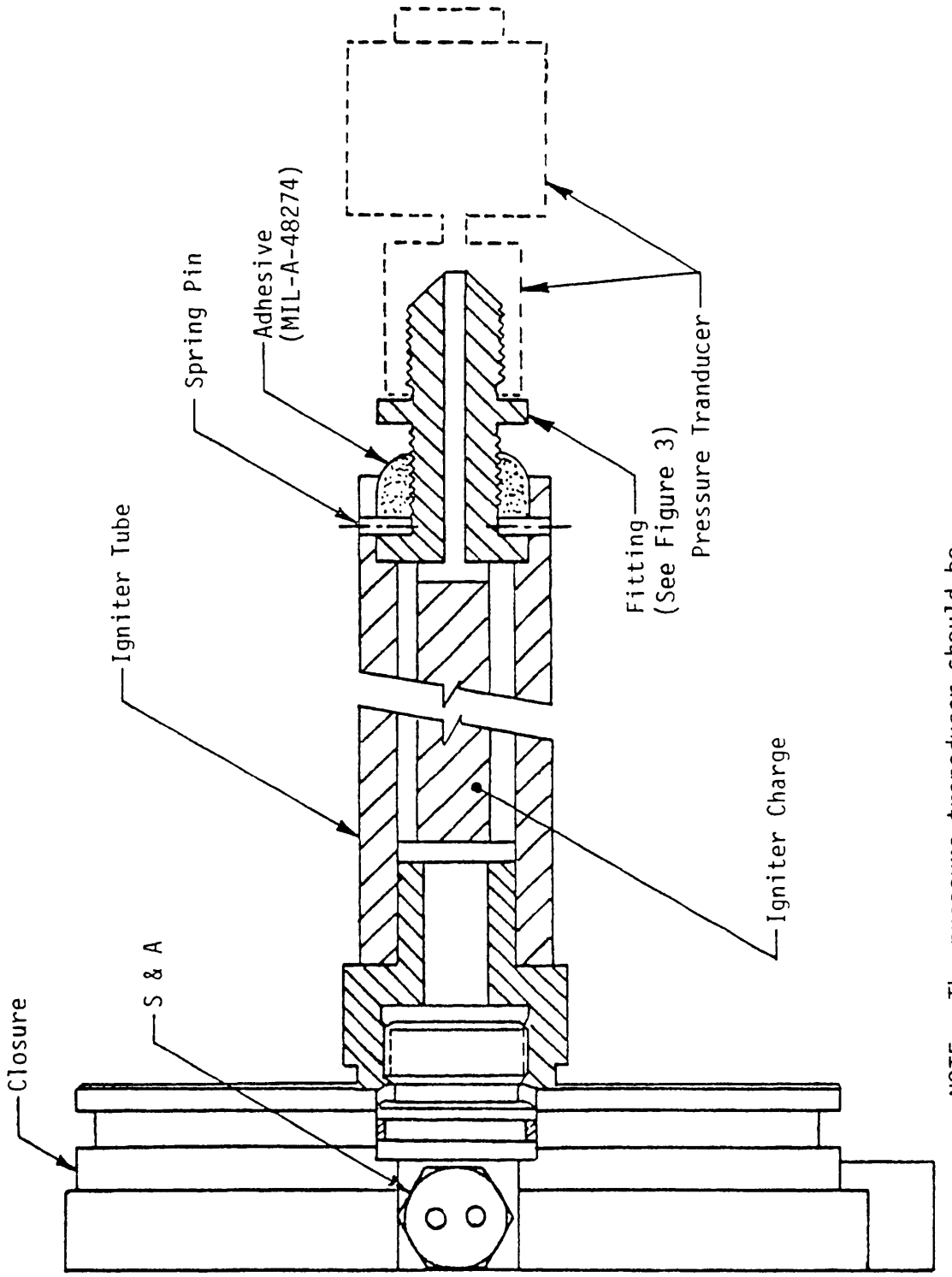


Figure 1. Pressure versus time function requirement.

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NOTE: The pressure transducer should be located as close to the fitting as practicable to minimize loss in pressure response.

Figure 2. Type II igniter test unit.

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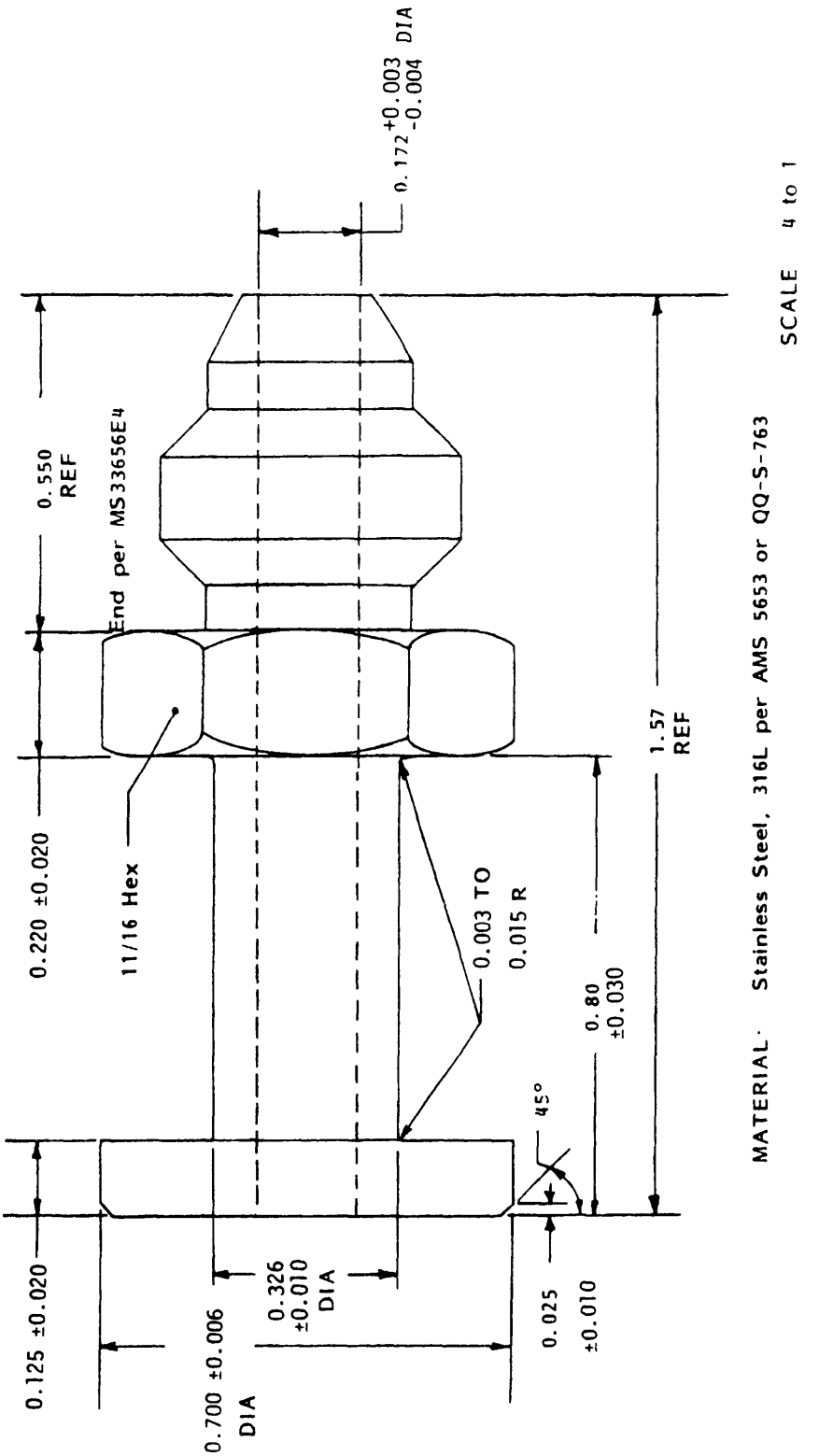


Figure 3. Igniter pressure transducer adapter.

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